<u>Tutorial and Lab Practice Seven: System</u>

Question 5: What is a signal? How can it be handled (describe three possible responses to a signal)? (1%) Where do signals come from (identify three sources)?

A small notification about an event to a running process, the CTRL + C key interrupts the current running program. This interruption occurs due to a kernel mechanism known as a signal, each signal has numerical code. Three responses to a signal consist of default response, signal handler which uses a function which activates on signal, and then the last response is to ignore the signal entirely. A signal can be sent from 3 different sources such as:

1: User/Keyboard (CTRL + C) = Interruption

- Using a SignInt associated with a number which can be used to stop process

2: Kernel

The kernel sends a signal to a process.

3: Another Process (Signal Handling)

- Default | Ignore | Function Call (handling)
- Receives a signal, and using a function redirect the signal to shut a process (which is handling).

Question 6: How does a process send a signal to another process? Provide three examples using (1%) different signals.

A Process will use a function in order to send a signal to another process, so this can be achieved by using system calls such as kill. Kill sends a signal to a process, the process which is sending the signal must have the same user ID as the target process. The process is able to send a signal to itself, a process may also send any signal to another process which includes signals coming from the keyboard, interval timers, or from the kernel.